

# TOWARDS AI-COMPLETE QUESTION ANSWERING: A SET OF PREREQUISITE TOY TASKS

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## ABSTRACT

One long-term goal of machine learning research is to produce methods that are applicable to reasoning and natural language, in particular building an intelligent dialogue agent. To measure progress towards that goal, we argue for the usefulness of a set of proxy tasks that evaluate reading comprehension via question answering. Our tasks measure understanding in several ways: whether a system is able to answer questions via chaining facts, simple induction, deduction and many more. The tasks are designed to be prerequisites for any system that aims to be capable of conversing with a human. We believe many existing learning systems can currently not solve them, and hence our aim is to classify these tasks into skill sets, so that researchers can identify (and then rectify) the failings of their systems. We also extend and improve the recently introduced Memory Networks model, and show it is able to solve some, but not all, of the tasks.

## 1 INTRODUCTION

There is a rich history of the use of synthetic tasks in machine learning, from the XOR problem which helped motivate neural networks (Minsky & Papert, 1969; Rumelhart et al., 1985), to circle and ring datasets that helped motivate some of the most well-known clustering and semi-supervised learning algorithms (Ng et al., 2002; Zhu et al., 2003), Mackey Glass equations for time series (Müller et al., 1997), and so on – in fact some of the well known UCI datasets (Bache & Lichman, 2013) are synthetic as well (e.g., *waveform*). Recent work continues this trend. For example, in the area of developing learning algorithms with a memory component synthetic datasets were used to help develop both the Neural Turing Machine of Graves et al. (2014) and the Memory Networks of Weston et al. (2014), the latter of which is relevant to this work.

One of the reasons for the interest in synthetic data is that it can be easier to develop new techniques using it. It is well known that working with large amounts of real data (“big data”) tends to lead researchers to simpler models as “simple models and a lot of data trump more elaborate models based on less data” (Halevy et al., 2009). For example,  $N$ -grams for language modeling work well relative to existing competing methods, but are far from being a model that truly understands text. As researchers we can become stuck in local minima in algorithm space; development of synthetic data is one way to try and break out of that.

In this work we propose a framework and a set of synthetic tasks for the goal of helping to develop learning algorithms for text understanding and reasoning. While it is relatively difficult to automatically evaluate the performance of an agent in general dialogue – a long term-goal of AI – it is relatively easy to evaluate responses to input questions, i.e., the task of question answering (QA). Question answering is incredibly broad: more or less any task one can think of can be cast into this setup. This enables us to propose a wide ranging set of different tasks, that test different capabilities of learning algorithms, under a common framework.

Our tasks are built with a unified underlying simulation of a physical world, akin to a classic text adventure game (Montfort, 2005) whereby actors move around manipulating objects and interacting

with each other. As the simulation runs, grounded text and question answer pairs are simultaneously generated. Our goal is to categorize *different kinds of questions* into skill sets, which become our tasks. Our hope is that the analysis of performance on these tasks will help expose weaknesses of current models and help motivate new algorithm designs that alleviate these weaknesses. We further envision this as a feedback loop where new tasks can then be designed in response, perhaps in an adversarial fashion, in order to break the new models.

The tasks we design are detailed in Section 3, and the simulation used to generate them in Section 4. In Section 5 we give benchmark results of standard methods on our tasks, and analyse their successes and failures. In order to exemplify the kind of feedback loop between algorithm development and task development we envision, in Section A we propose a set of improvements to the recent Memory Network method, which has shown to give promising performance in QA. We show our proposed approach does indeed give improved performance on some tasks, but is still unable to solve some of them, which we consider as open problems.

## 2 RELATED WORK

Several projects targeting language understanding using QA-based strategies have recently emerged. Unlike tasks like dialogue or summarization, QA is easy to evaluate (especially in true/false or multiple choice scenarios) and hence makes it an appealing research avenue. The difficulty lies in the definition of questions: they must be unambiguously answerable by adult humans (or children), but still require some thinking. The *Allen Institute for AI*'s flagship project ARISTO<sup>1</sup> is organized around a collection of QA tasks derived from increasingly difficult science exams, at the 4th, 8th, and 12th grade levels. Richardson et al. (2013) proposed the *MCTest*<sup>2</sup> a set of 660 stories and associated questions intended for research on the machine comprehension of text. Each question requires the reader to understand different aspects of the story.

These two initiatives go in a promising direction but interpreting the results on these benchmarks remain complicated. Indeed, no system has yet been able to fully solve the proposed tasks and since many sub-tasks need to be solved to answer any of their questions (coreference, deduction, use of common-sense, etc.), it is difficult to clearly identify capabilities and limitations of these systems and hence to propose improvements and modifications. As a result, conclusions drawn from these projects are not much clearer than that coming from more traditional works on QA over large-scale Knowledge Bases (Berant et al., 2013; Fader et al., 2014). Besides, the best performing systems are based on hand-crafted patterns and features, and/or statistics acquired on very large corpora. It is difficult to argue that such systems actually understand language and are not simply light upgrades of traditional information extraction methods (Yao et al., 2014). The system of Berant et al. (2014) is more evolved since it builds a structured representation of a text and of a question to answer. Despite its potential this method remains highly domain specific and relies on a lot of prior knowledge.

Based on these observations, we chose to conceive a collection of much simpler QA tasks, with the main objective that failure or success of a system on any of them can unequivocally provide feedback on its capabilities. In that, we are close to the *Winograd Schema Challenge* Levesque et al. (2011), which is organized around simple statements followed by a single binary choice question such as: “*Joan made sure to thank Susan for all the help she had received. Who had received the help? Joan or Susan?*”. In this challenge, and our tasks, it is straightforward to interpret results. Yet, where the Winograd Challenge is mostly centered around evaluating if systems can acquire and make use of background knowledge that is not expressed in the words of the statement, our tasks are self-contained and are more diverse. By self-contained we mean our tasks come with both training data and evaluation data, rather than just the latter as in the case of ARISTO and the Winograd Challenge. *MCTest* has a train/test split but the training set is likely too small to capture all the reasoning needed to do well on the test set. In our setup one can assess the amount of training examples needed to perform well (which can be increased as desired) and commonsense knowledge and reasoning required for the test set should be contained in the training set. In terms of diversity, some of our tasks are related to existing setups but we also propose many additional ones; tasks 8 and 9 are inspired by previous work on lambda dependency-based compositional semantics (Liang et al., 2013; Liang, 2013) for instance. For us, each task checks one skill that the system must

<sup>1</sup><http://allenai.org/aristo.html>

<sup>2</sup><http://research.microsoft.com/mct>

have and we postulate that performing well on all of them is a prerequisite for any system aiming at full text understanding and reasoning.

### 3 THE TASKS

**Principles** Our main idea is to provide a set of tasks, in a similar way to how software testing is built in computer science. Ideally each task is a “leaf” test case, as independent from others as possible, and tests in the simplest way possible one aspect of intended behavior. Subsequent (“non-leaf”) tests can build on these by testing combinations as well. The tasks are publicly available at <http://fb.ai/babi>. Source code to generate the tasks is available at <https://github.com/facebook/bAbI-tasks>.

Each task provides a set of training and test data, with the intention that a successful model performs well on test data. Following Weston et al. (2014), the supervision in the training set is given by the true answers to questions, and the set of *relevant* statements for answering a given question, which may or may not be used by the learner. We set up the tasks so that correct answers are limited to a single word (Q: *Where is Mark?* A: *bathroom*), or else a list of words (Q: *What is Mark holding?*) as evaluation is then clear-cut, and is measured simply as right or wrong.

All of the tasks are noiseless and a human able to read that language can potentially achieve 100% accuracy. We tried to choose tasks that are natural to a human: they are based on simple usual situations and no background in areas such as formal semantics, machine learning, logic or knowledge representation is required for an adult to solve them.

The data itself is produced using a simple simulation of characters and objects moving around and interacting in locations, described in Section 4. The simulation allows us to generate data in many different scenarios where the true labels are known by grounding to the simulation. For each task, we describe it by giving a small sample of the dataset including statements, questions and the true labels (in red) in Tables 1 and 2.

**Single Supporting Fact** Task 1 consists of questions where a previously given single supporting fact, potentially amongst a set of other irrelevant facts, provides the answer. We first test one of the simplest cases of this, by asking for the location of a person, e.g. “*Mary travelled to the office. Where is Mary?*”. This kind of task was already employed in Weston et al. (2014). It can be considered the simplest case of some real world QA datasets such as in Fader et al. (2013).

**Two or Three Supporting Facts** A harder task is to answer questions where two supporting statements have to be chained to answer the question, as in task 2, where to answer the question “*Where is the football?*” one has to combine information from two sentences “*John is in the playground*” and “*John picked up the football*”. Again, this kind of task was already used in Weston et al. (2014). Similarly, one can make a task with three supporting facts, given in task 3, whereby the first three statements are all required to answer the question “*Where was the apple before the kitchen?*”.

**Two or Three Argument Relations** To answer questions the ability to differentiate and recognize subjects and objects is crucial. In task 4 we consider the extreme case where sentences feature re-ordered words, i.e. a bag-of-words will not work. For example, the questions “*What is north of the bedroom?*” and “*What is the bedroom north of?*” have exactly the same words, but a different order, with different answers. A step further, sometimes one needs to differentiate three separate arguments. Task 5 involves statements like “*Jeff was given the milk by Bill*” and then queries who is the giver, receiver or which object is involved.

**Yes/No Questions** Task 6 tests, on some of the simplest questions possible (specifically, ones with a single supporting fact), the ability of a model to answer true/false type questions like “*Is John in the playground?*”.

**Counting and Lists/Sets** Task 7 tests the ability of the QA system to perform simple counting operations, by asking about the number of objects with a certain property, e.g. “*How many objects is Daniel holding?*”. Similarly, task 8 tests the ability to produce a set of single word answers in the form of a list, e.g. “*What is Daniel holding?*”. These tasks can be seen as QA tasks related to basic database search operations.